

Online Dating and Identifying Falsified Heights

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Introduction

A large demographic of US adults use dating apps. In general, 3 out of every 10 adults use dating apps. For people under 30, this amount increases to 50% (Vogels & McClain 2023). As the popularity of apps like Tinder, Hinge, and Bumble, increases, a growing problem is emerging: people have a tendency to lie on their profiles to make them seem more desirable. Specifically, many users inflate their reported height in an attempt to appear more attractive and get more matches. According to a GQ Modern Love survey of 604 men ages 16-44, 27% of them misreporting their height, with a majority of that group adding about two inches to their reported height.

The question is, on dating platforms where height is often self-reported and inflated, how can we verify this information? One source of truth that could be used is the images on the profiles. A majority of dating apps have images of the user they keep on a profile in order to show them to potential matches. Assuming these images are unaltered and are actually of the purported user, the signals in these photos could potentially be used to estimate the user's height.

Prior work on height estimation from images has explored the use of deep learning models trained on full-body image datasets. The Celeb-FBI dataset provides labeled full-body celebrity images with height, weight, age, and gender information, making it relevant for supervised height prediction. In addition, pretrained vision models such as DINOv2 provide general-purpose visual representations that can be adapted to downstream tasks with limited task-specific data. This project builds on these resources by fine-tuning DINOv2 for height regression and comparing its performance against simpler baseline models that can estimate the height of a person from a photograph.

Approach

This project treats height estimation from an image as a supervised regression task. The dataset used for this task was the Celeb-FBI face/body height dataset. Since the goal of the model was to predict a continuous height value, entries without usable height labels were removed before training. The remaining data was split into training, validation, and test sets using an 80/10/10 split. During model training, images were resized to 224×224 pixels and normalized using the standard ImageNet mean and standard deviation before being passed into the models.

The main model architecture used facebook/dinov2-base as a pretrained visual backbone. DINOv2 was selected because it has already learned general-purpose image representations, making it a strong candidate for a task with limited domain-specific training data. For height prediction, the model uses the DINOv2 CLS token as the visual representation of the image and passes it into a small regression head.

The regression head consists of a Linear(768, 256) layer, a ReLU activation, dropout with probability 0.2, and a final Linear(256, 1) layer that outputs a single predicted height value in centimeters.

To understand whether the DINOv2 model was improving on simpler approaches, two baselines were implemented. The first was a mean-height baseline, which always predicts the average height from the training data. This served as an absolute baseline because any useful image-based model should perform better than simply predicting the same value for every person. The second was a small convolutional neural network trained from scratch. This CNN baseline was included to test whether a simpler model could learn useful height features directly from the dataset without relying on pretrained representations. The CNN used a lightweight convolutional architecture with two convolutional layers, max pooling, dropout, and fully connected layers that output a single predicted height value.

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SimpleCNN architecture:

SimpleCNN(
  (conv1): Conv2d(3, 16, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
  (conv2): Conv2d(16, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
  (pool): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1, ceil_mode=False)
  (fc1): Linear(in_features=100352, out_features=64, bias=True)
  (fc2): Linear(in_features=64, out_features=1, bias=True)
  (relu): ReLU()
  (dropout): Dropout(p=0.5, inplace=False)
)

Layer flow:
- Input: output 3 x 224 x 224 | RGB image
- Conv1 + ReLU: output 16 x 224 x 224 | 3x3, 16 filters
- MaxPool1: output 16 x 112 x 112 | 2x2
- Conv2 + ReLU: output 32 x 112 x 112 | 3x3, 32 filters
- MaxPool2: output 32 x 56 x 56 | 2x2
- Flatten: output 100352 | 32 x 56 x 56
- FC1 + ReLU + Dropout: output 64 | 100352 -> 64, p=0.5
- FC2: output 1 | 64 -> 1
- Output: output Predicted height | Scalar regression
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Figure 1: CNN baseline architecture.

The DINOv2 model was tested in multiple forms. In the first version, the DINOv2 backbone was kept frozen and only the regression head was trained. This tested whether the pretrained image features already contained enough information to support height prediction. In the second version, the DINOv2 backbone was fine-tuned along with the regression head so the model could adapt its visual representations to the specific task. Fine-tuning used differential learning rates, with a smaller learning rate of 1e-5 for the pretrained backbone and a larger learning rate of 1e-4 for the regression head.

The project also included a data augmentation strategy designed to make the model more robust to images that are not full-body photos. The augmentation process randomly blackens the lower portion of some training images, covering approximately 20–60% of the image. This was meant to simulate real profile photos where part of the body may be blocked by objects such as desks, counters, furniture, or image cropping. This augmentation was applied only to the training data, while validation and test images were left unchanged so model performance could be evaluated on the original data distribution.

The implementation was organized into separate components for data cleaning, augmentation, baseline modeling, DINOv2 training, fine-tuning, error analysis, and user-facing inference. This separation made the project easier to reproduce and made it possible to compare each modeling stage independently. Data preparation and augmentation were handled separately from model training, while the baseline models and DINOv2 experiments were kept in distinct scripts and folders.

To demonstrate the model in a user-facing format, the repository also includes a Streamlit application. The app allows a user to upload a photo and receive a predicted height in both centimeters and feet/inches. It loads the saved DINOv2 checkpoint, applies the same resizing and normalization steps used during training, and returns the predicted height with an approximate range. The application was designed as a lightweight demonstration of the final model rather than as part of the formal evaluation pipeline. The initial Streamlit app code was generated with assistance from OpenAI Codex and then reviewed, edited, and integrated into the project.

Evaluation

The evaluation strategy for this project was designed to compare whether increasingly complex modeling approaches improved height prediction from images. All models were evaluated using the same cleaned dataset split so that performance differences could be attributed to the modeling approach rather than differences in the data. The models compared included a mean-height baseline, a convolutional neural network trained from scratch, a frozen DINOv2 model with only the regression head trained, a fine-tuned DINOv2 model, and a fine-tuned DINOv2 model trained with augmented data.

The mean-height baseline was included as the absolute baseline for the task. Since this model always predicts the average height from the training data, it does not use image information. Any useful image-based model should perform better than this baseline. The CNN baseline was included to test whether a smaller model trained from scratch could learn meaningful height-related features directly from the dataset. The frozen DINOv2 model evaluated whether pretrained visual embeddings alone were sufficient for height estimation, while the fine-tuned DINOv2 models tested whether adapting the pretrained backbone to the Celeb-FBI dataset improved performance.

Because height prediction is a regression problem, the primary metric used for evaluation was mean squared error (MSE). MSE was used because it penalizes larger prediction errors more heavily than smaller ones, which is important in this context because a large height error would make the model unreliable for user-facing prediction. In addition to MSE, the final model analysis also considered signed error and bias by true height range. This helped show whether the model was simply making random errors or whether it systematically overpredicted or underpredicted certain groups of people.

Model selection was based on validation performance rather than training performance. This was important because lower training error does not necessarily mean that the model generalizes better to new images. During fine-tuning, training and validation MSE were tracked across epochs to identify whether additional training improved or worsened performance on unseen validation data. The best-performing model was then evaluated on the held-out test set.

The project also included qualitative error analysis for the strongest model. Since the augmented fine-tuned DINOv2 model produced the best overall test performance, the deeper error analysis focused on that version rather than every model. This analysis examined actual versus predicted height, as well as signed prediction error across true-height bins. These visualizations were used to determine whether the model performed consistently across the full height distribution or whether it was more reliable for certain height ranges than others.

Overall, the evaluation framework tested both model accuracy and model behavior. The quantitative metrics allowed direct comparison between the baseline models and the DINOv2 variants, while the qualitative analysis provided a more detailed view of where the best-performing model succeeded and failed. This distinction was important because a model can achieve strong aggregate performance while still showing systematic errors for unusually short or unusually tall individuals.

Results and Discussion

For all test-set comparisons, we used mean squared error (MSE) to evaluate model performance. For our mean baseline, the prediction was 171.42 for every image, and we got a test accuracy of 141.11. The convolutional neural network, the performance decreased drastically, with a MSE of 287.43.

For the base DinoV2 model, the model performed better than the CNN baseline and worse than the mean baseline, with a MSE of 220.82. With finetuning, the MSE improved to 134.64. Furthermore the model that was finetuned on the augmented data performed the best overall, with a MSE of 127.84. These accuracies are summarized below in **Figure 2**.

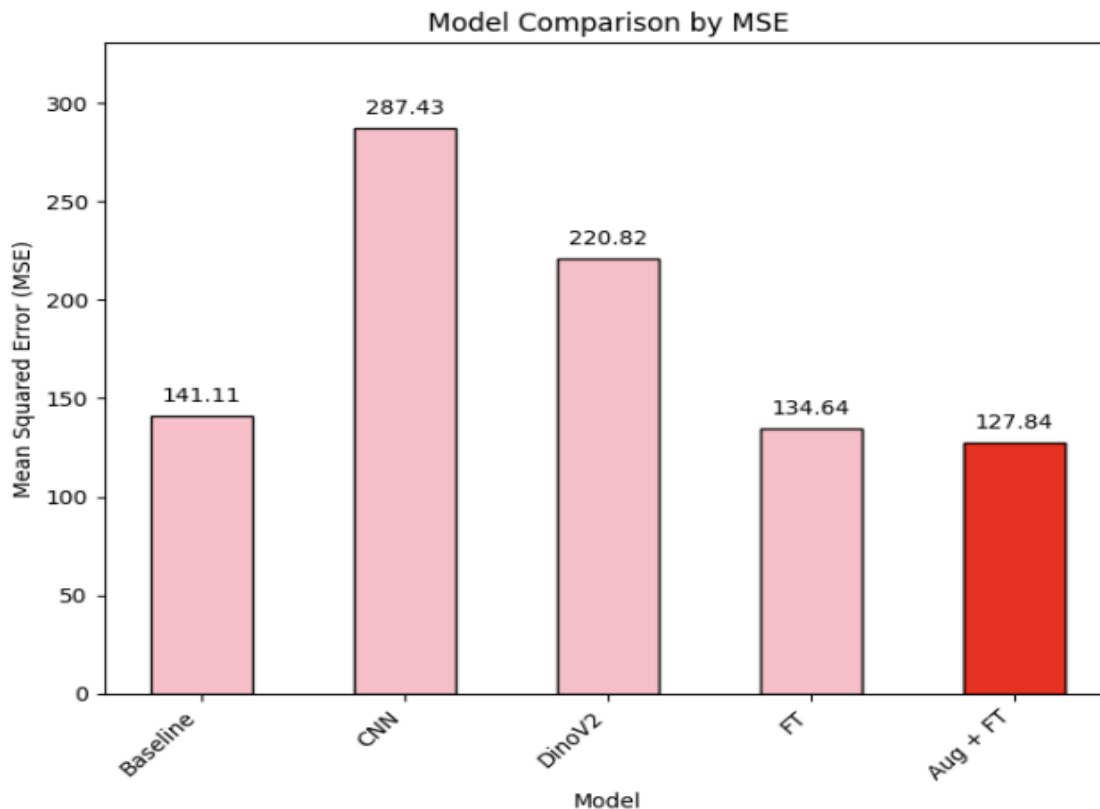


Figure 2: MSE of all trained models on Test Set

These results suggest that fine-tuning was necessary for DINOv2 to perform well on this task. When only the regression head was trained, the model did not appear to extract enough height-relevant information from the image representation to outperform the mean baseline. After fine-tuning, however, the model was better able to adapt its visual features to the height prediction task, producing substantially lower MSE. This indicates that the pretrained representation was useful, but it needed task-specific adaptation. The MSE values also show that while data augmentation did help performance, it did not do so substantially. This suggests that another form of augmentation could be done to make the model more robust.

Because the augmented fine-tuned DINOv2 model was the best-performing model overall, we conducted additional error analysis on this version to better understand where the model was succeeding and failing. As shown in **Figure 3**, the model's predictions generally follow the correct direction: taller individuals tend to receive higher predicted heights. However, the best-fit line is much flatter than the ideal $y = x$ line, which suggests that the model compresses predictions toward the center of the height distribution. In other words, it tends to overpredict shorter individuals and underpredict taller individuals rather than fully capturing the extremes.

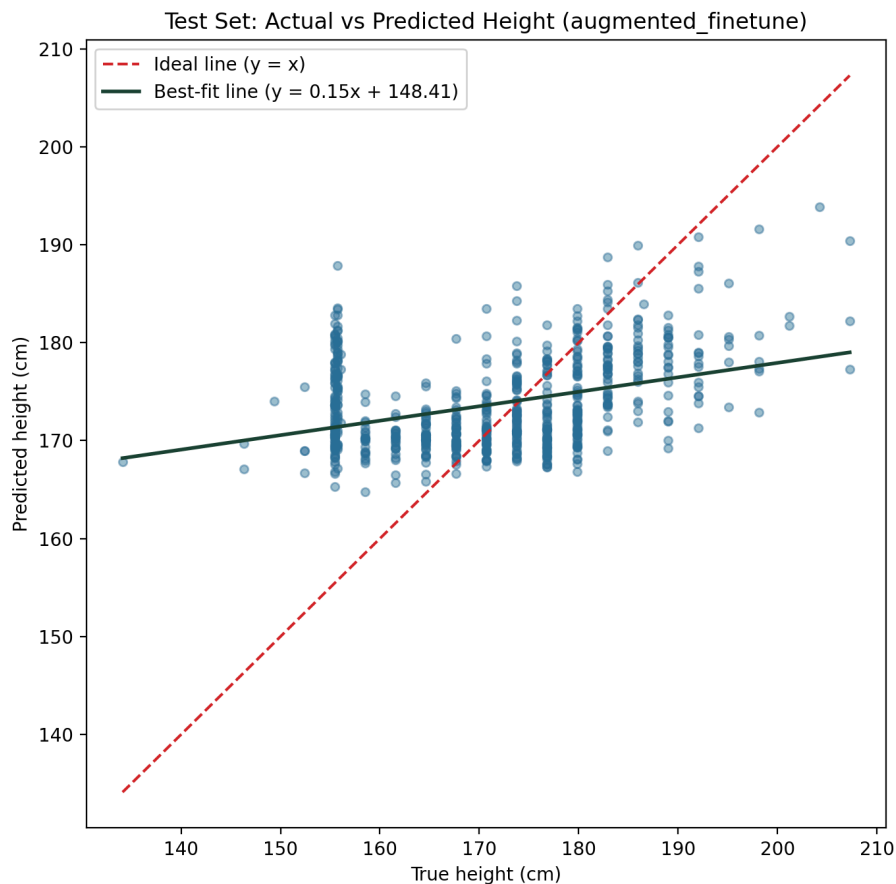


Figure 3: Test Set Actual vs. Predicted Height for the Augmented Fine-Tuned Model

This pattern is also visible when signed error is grouped by true height bin, as shown in Figure 4. For individuals under 150 cm, the model overpredicts height by an average of 25.6 cm, and for individuals in the 150–159 cm range, it overpredicts by 17.6 cm. The bias becomes smaller near the middle of the distribution, with an average error of only -2.8 cm for the 170–179 cm bin. However, the direction reverses for taller individuals: the model underpredicts by 14.5 cm for the 190–199 cm bin and by 20.0 cm for the 200+ cm bin. This suggests that the model performs best for individuals closer to the center of the training distribution, while predictions become less reliable for unusually short or unusually tall subjects.

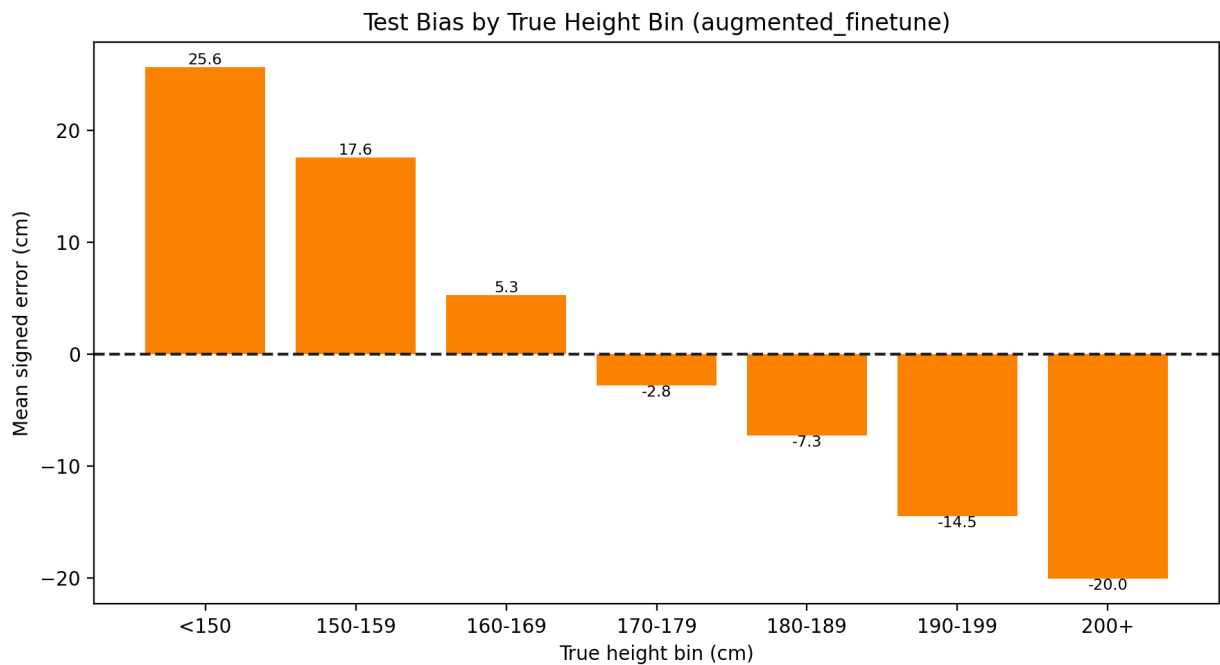


Figure 4: Test Bias by True Height Bin for the Augmented Fine-Tuned Model

Overall, this error pattern is important because it shows that the model’s aggregate MSE does not fully capture how performance varies across different types of users. Although the augmented fine-tuned model achieved the best overall MSE, it still shows systematic bias at the extremes of the height range. This limitation is especially relevant for a dating-app context, because the model could unfairly flag shorter or taller users as misreporting their height when the error is actually caused by the model’s tendency to regress predictions toward average heights.

Also as part of our experimentation, we varied the amount of fine tuning on the base DinoV2 model (without data augmentation). We found that three epochs yielded the best results. The number of training epochs with their corresponding validation MSEs are shown in **Table 1** below.

Epoch	Training MSE	Validation MSE
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1	223.41	147.79
2	183.89	130.32
3	172.86	135.03
4	156.01	154.26
5	134.73	137.38

Table 1: MSE of fine-tuned model on given epoch

The increase in validation MSE after epoch three could be explained by catastrophic forgetting occurring during the fine tuning of the model. The DinoV2 model is trained to identify the features within an image. It likely started to lose the ability to perform that task as it was being fine-tuned, overall decreasing the performance of the model.

Ethics and Limitations

Many ethical concerns arise at the prospect of implementing this into an actual dating platform. First, there is the concern that the model may miscalculate a person's height. Because there is always a margin of error with these types of vision models, the model may falsely state that someone is lying about their height or falsely say that a person is their reported height. If the purpose of this model is to have some user facing element, there needs to be consideration as to how it deals with perceived false heights. If this information is displayed on someone's profile, it could potentially be implemented with some form of appeal process for users who believe the model is incorrect. Alternatively, the model could be used when a user reports an account of misrepresenting their height, and be used as a factor in considering whether it is valid or not.

The implementation of this model also raises the concern of user privacy. There are many people who may not want their likeness used in the training of subsequent models. Furthermore some may not want their pictures being evaluated on either by the model. By the nature of this model being used to figure out deception, people may not want it used on them.

The primary limitation of this model comes from the data that was used in training. The Celeb-FBI dataset is composed of adults over the age of 21, who are taller than 4'8" and shorter than 6' 5". The model may struggle in real world applications that are outside of this domain. Because of this lack of data, the model could not be evaluated on this subset of tasks. Further work focusing on making the estimate more robust, could include an expanded dataset to account for this limitation.

Another factor that may influence model performance is the angle that the picture was taken. The photograph may be taken in a misleading way in order to increase the perceived height of the subject. Our model does not explicitly account for this. The photos in the Celeb-FBI set are primarily images of the subject facing forward, with the image being taken directly in front of them. Although data augmentation

could account for this (e.g., modifying image proportions to adjust perceived height), implementing and testing this approach fell outside the scope and timeline of our current research.

Future Work

Future work could possibly explore other predictable factors, such as age, that people may falsify on a dating app. These are also metrics that people lie about on dating apps in order to seem more attractive. The Celeb-FBI dataset has several such values that could be incorporated in future prediction tasks.

Given time constraints, this work only explored the fine tuning of the DinoV2 model. More analysis could be done on different base models and testing which perform with the highest accuracy. Furthermore we did not vary the amount of training on the regression head. Future works could explore how the training of these weights influences the overall accuracy on the given prediction task.

This work explored data augmentation as a means of making the model more robust to photos that are not full body photos. However only one form of augmentation (i.e., blacking out a randomly placed rectangle in the bottom of the image) was explored due to time limitations and dataset restrictions. Future work could incorporate other methods of data augmentation, such as removing the bottom half of the image or rotating the photo. Alternatively, work could focus on applying this task to a different category of photos, such as photos taken from an angle or people being seated.

Team contribution

Angelo came up with the idea for the model and wrote a large portion of the project abstract. Azalea looked into models that could be fine-tuned for this task and found the DinoV2 model. Angelo primarily worked on finding, cleaning, and augmenting the data and evaluating the various forms of the DinoV2 model (base and fine-tuned). Azalea worked on the implementation of the baseline evaluations, calculating the mean baseline, as well as the creation of the convolutional neural network. She also created the user interface for evaluating images. Both Angelo and Azalea worked on the presentation slide deck.

References

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